External-Space Self-Consistency Law

A speculative theoretical paper on a hypothetical temporal quantum machine

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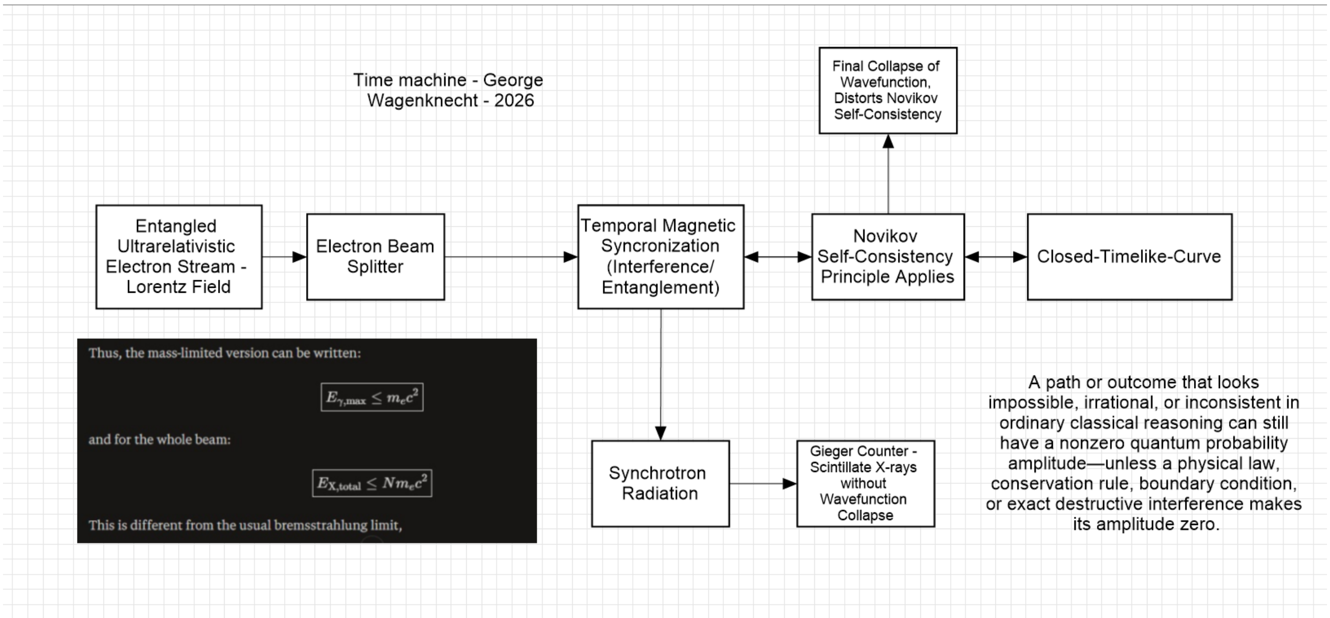

Figure 1. Conceptual architecture of the proposed machine.

Abstract

This paper proposes a speculative principle called the External-Space Self-Consistency Law. The principle describes a hypothetical machine in which possible configurations of an external physical space are represented by quantum amplitudes, while a global consistency condition determines which configurations can retain nonzero amplitude. The machine is not assumed to rewrite an already established past. Instead, configurations that cannot satisfy the complete spacetime constraints are suppressed or excluded. The mathematical model below is explicitly speculative and is not presented as an experimentally established law of physics.

1. The Proposed Machine

The machine is treated as a conceptual device rather than a demonstrated time machine. Its imagined architecture contains an electron-beam stage, temporal magnetic synchronization, a hypothetical closed-timelike-curve constraint, radiation, and a final measurement stage. The defining theoretical feature is the proposed global constraint between the machine and an external configuration.

2. External Space as an Amplitude Space

Let $|E_i\rangle$ denote a possible configuration of external space. A general state can be represented as:

$$|\Psi\rangle = \Sigma_i A_i |E_i\rangle$$

Here $A_i \in \mathbb{C}$ is the complex amplitude associated with external configuration $|E_i\rangle$, and $|A_i|^2$ gives its probability weight when the $\{|E_i\rangle\}$ form the relevant measurement basis.

3. External-Space Self-Consistency Law

Law: A physically realizable external configuration must possess a nonzero amplitude under its complete spacetime constraints; configurations that cannot consistently close upon themselves are excluded rather than retroactively altered.

$$\hat{C}|E_i\rangle = |E_i\rangle \text{ for a self-consistent configuration}$$

$$\hat{C}|E_i\rangle = 0 \text{ for an inconsistent configuration}$$

The idealized surviving state is then a projection of the candidate state into the self-consistent subspace:

$$|\Psi_{\text{allo}}w_{e}d\rangle = \hat{C}|\Psi\rangle \,/\, \sqrt{\langle\Psi|\hat{C}|\Psi\rangle}$$

4. Why the Past Need Not Change

A literal retroactive rewrite would require the state at an earlier time to be transformed after the fact:

$$|\Psi(t_{\text{past}})\rangle \to |\Psi'(t_{\text{past}})\rangle$$

The proposed law instead constrains complete external configurations. If a candidate configuration cannot satisfy the global condition, its amplitude is suppressed. Thus the model does not require an already observed past to be replaced by a second past.

5. Global Constraint

Let F(E) encode the complete temporal, dynamical, and boundary constraints on an external configuration E. The proposed condition can be written:

$$A(E) \neq 0 \Longrightarrow F(E) = 0$$

This says that nonzero amplitude requires global consistency. It is a toy constraint equation, not a derivation from established quantum field theory.

6. Relation to Closed Timelike Curves

The proposal is conceptually related to self-consistency ideas in closed-timelike-curve physics. Novikov's self-consistency principle requires physical solutions in a spacetime containing closed timelike curves to remain globally consistent. Deutsch's quantum formulation instead uses a density-matrix fixed-point condition. Postselection approaches provide another mathematical formulation. These frameworks are distinct; the present paper combines their broad conceptual intuition into a speculative external-space amplitude model.

7. Machine-Level Interpretation

The electron-beam stage produces candidate spatial alternatives. Temporal synchronization defines the machine's relevant phase and timing constraints. The hypothetical consistency region imposes $\hat{C}$. Radiation and X-ray detection provide possible measurement channels. The final detector therefore samples the external configuration after the proposed consistency restriction.

8. Proposed Experimental Signature

If the hypothesized mechanism existed, its distinguishing signature would not simply be a visible change to the past. Instead, repeated measurements might show that certain externally defined configurations have systematically suppressed probabilities under controlled temporal boundary conditions. A genuine test would need to separate such an effect from ordinary interference, decoherence, detector bias, causal correlations, and postselection artifacts.

9. Limitations

No established experiment demonstrates a macroscopic closed timelike curve or a device capable of selecting external reality through a temporal consistency operator. The equations in this paper therefore define a conceptual hypothesis. In particular, assigning $\hat{C}$ to a real physical mechanism is an open problem, and the existence of such an operator in nature does not follow from the notation alone.

10. Conclusion

The central proposal is a shift from changing histories to filtering external configurations. Quantum amplitudes describe candidate external states, while a hypothetical global consistency condition determines which candidates remain physically realizable. Under this interpretation, the machine does not rewrite the past; rather, only externally consistent configurations survive as complete physical solutions.

Author's Note

The External-Space Self-Consistency Law is introduced by George Wagenknecht as a speculative conceptual law for the proposed machine. It is not claimed here as an established law of physics.